

What the CPU CFT-Extraction Experiments Actually Establish

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This note audits the completed CPU experiments in `cpu.cft.extraction` from the formal target through the numerical implementation, saved arrays, fits, figures, and convergence diagnostics. The calculation sought the effective central charge from the finite-size correction to a trajectory free-energy density and typical scaling dimensions from tangent-cocycle and Choi-transfer gaps. The pipeline successfully measures a slow interface channel and, at moderate time, the two independent Gaussian spectral routes give a mutually consistent candidate scale near $x \simeq 0.19$. The experiments do *not* yet determine c_{eff} or an asymptotic scaling dimension. The fitted central charge changes from -345 at $T/L = 2$ to -159 at $T/L = 5$ and $+40$ at $T/L = 20$, while the underlying free-energy curves retain measurable temporal drift. At $T/L = 20$ the Choi eigenvalues have saturated to the numerical endpoints ± 1 , eliminating every final-time finite rapidity. The remaining long-time tangent estimate is strongly dependent on whether the finite-size fit is allowed a nonzero intercept. These are convergence and numerical dynamic-range failures, not evidence for the large quoted central charges. The main positive outcome is therefore methodological: the observable pipeline works and exposes a structured low-lying transfer spectrum, but a controlled CFT extraction requires seeded reproducibility, late-time increment estimators, higher statistics, sector-resolved gaps, and a log-stable Choi formulation.

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I. SCOPE AND BOTTOM LINE

The evidentiary boundary is deliberately conservative. This note uses only campaigns for which the driver wrote the complete scalar table and all per-size trajectory, tangent, and Choi files. It includes

1. $N_x = 20$, $L = N_y = 20, 30, 40$, $T = 2L$, ten samples per size;
2. $N_x = 20$, $L = 20, 30, 40$, $T = 5L$, ten samples per size, formed by combining the separately completed $L = 20$ and $L = 30, 40$ runs with identical declared parameters; and
3. $N_x = 20$, $L = 20, 22, 24$, $T = 20L$, ten samples per size.

The $T = 20L$, $L = 26, 28, 30$ extension was still running when this note was prepared. It is not used in a fit, table, trend, or extrapolation. In particular, the completed $T = 5L$ and the unfinished $T = 20L$ realizations at $L = 30$ are never mixed.

The conclusions, ordered from strongest to weakest, are:

1. The implemented experiment detects a slow, size-dependent interface transfer mode. The tangent and Choi routes track one another closely while both are numerically resolved.
2. At $T/L = 5$, direct and through-origin gap estimates from both routes cluster near $x \simeq 0.19$. This is a useful preasymptotic candidate, not a measured universal exponent.
3. The central-charge extraction is uncontrolled. Its sign and magnitude change with the run time, and the expected $O(L^{-2})$ signal is much smaller than the present statistical and temporal systematics.
4. The late-time Choi failure is understood: finite rapidities are exponentially mapped to eigenvalues indistinguishable from ± 1 under the fixed endpoint tolerance. Endpoint counts consequently mix physical endpoint modes with finite-precision saturation.
5. The rank scan reveals a smooth additive low-energy tower and striking tangent–Choi agreement through the range where the Choi data remain supported. Since it is a postprocessing sum of one-body order statistics at $T/L = 2$, it is not yet a sector-resolved CFT spectrum.

II. FORMAL TARGET

A. Trajectory weights and the cylinder free energy

For a measurement record $\xi = (m_1, \dots, m_t)$ sampled with the Born rule, the normalized trajectory probability is

$$p_\xi = \prod_{k=1}^t p(m_k \mid m_1, \dots, m_{k-1}), \quad Y_\xi = -\log p_\xi. \quad (1)$$

The code accumulates exactly this Y_ξ for the physical measurement sequence used here; the perfect-correction factor is unity, so its logarithm vanishes. With circumference L and the conversion between circuit cycles and isotropic Euclidean time denoted by α , the finite-time estimator is

$$f_0(L, t) = \frac{1}{\alpha L t} \mathbb{E}_\xi[Y_\xi(t)]. \quad (2)$$

If the long-distance replicated transfer problem is governed by a conformal fixed point on a cylinder, its stationary free-energy density has the expansion

$$f_0(L) = f_\infty - \frac{\pi c_{\text{eff}}}{6L^2} + \frac{b_4}{L^4} + \dots. \quad (3)$$

Thus a straight-line fit of f_0 versus L^{-2} has slope $-\pi c_{\text{eff}}/6$. This statement presupposes the limits $t/L \rightarrow \infty$ and $L \rightarrow \infty$, a fixed and calibrated α , and sufficiently small irrelevant and boundary corrections. A numerically precise line through three transient points is not by itself evidence that these conditions hold.

There is also an estimator issue worth separating from physics. The production output uses the cumulative value in Eq. (2) at the final cycle. If initialization contributes an $O(L)$ boundary term B_ξ , then

$$\frac{Y_\xi(t)}{Lt} = f_0(L) + \frac{B_\xi}{Lt} + \dots, \quad (4)$$

so convergence is only algebraic in time. A late-time slope of Y_ξ/L , or block increments $[Y(t_2) - Y(t_1)]/[L(t_2 - t_1)]$, removes the leading additive boundary term and is the more appropriate stationary estimator.

B. Typical transfer gaps and scaling dimensions

Let $S_A(\xi, t)$ denote a singular value or sector amplitude of the trajectory transfer operator, with S_0 the dominant reference. The quenched or typical gap is formally

$$\mu_A = - \lim_{t \rightarrow \infty} \frac{1}{t} \mathbb{E}_\xi \log \left| \frac{S_A(\xi, t)}{S_0(\xi, t)} \right|. \quad (5)$$

At a conformal fixed point,

$$\mu_A(L) = \frac{2\pi\alpha}{L} x_A^{\text{typ}} + o(L^{-1}), \quad \frac{\mu_A(L)}{\alpha L} = \frac{2\pi x_A^{\text{typ}}}{L^2} + \dots. \quad (6)$$

The production fit therefore regresses $\mu/(\alpha L)$ against L^{-2} and divides the slope by 2π . Allowing a free intercept is a diagnostic for corrections or a residual noncritical gap. With only three nearby sizes, however, slope and intercept can trade strongly; a through-origin fit provides an important sensitivity check rather than an alternative theorem.

The formal quenched average in Eq. (5) is a mean of trajectory logarithms. The scripts instead take the median of the per-trajectory smallest finite rates and bootstrap that median. This robust statistic is sensible in a heavy-tailed sample of ten, but it is not exactly the formal estimator. The distinction must remain explicit until the mean, median, and tail behavior are compared at substantially higher statistics.

C. Tangent cocycle

Linearizing the Gaussian covariance update about a physical trajectory defines a random tangent cocycle. QR stabilization accumulates logarithmic stretch factors and produces finite-time Lyapunov rates $\lambda_j(t)$. The implemented one-body proxy is

$$\Delta_{\text{tan}}(t) = \min_j |\lambda_j(t)|. \quad (7)$$

Interpreting Eq. (7) as a CFT transfer gap assumes that the neutral or dominant reference exponent is exactly zero and that the corresponding one-quasiparticle excitation is permitted in the target sector. More generally the physical quantity is a difference $|\lambda_j - \lambda_0|$ with charge and parity constraints. The QR code also clips null directions to large finite negative logarithms; the saved null count is cumulative over QR events, not the multiplicity of final modes removed from a spectrum.

D. Choi/Kraus transfer spectrum

The second construction contracts the cumulative Gaussian channel into a reduced Choi covariance. Its canonical eigenvalues occur as $\pm a_j$ with $|a_j| \leq 1$. Finite single-particle rapidities are reconstructed as

$$\epsilon_j(t) = \frac{|\text{artanh } a_j(t)|}{t}, \quad \Delta_{\text{choi}}(t) = \min_j \epsilon_j(t). \quad (8)$$

The implementation calls a_j an endpoint when $||a_j| - 1| < 10^{-8}$ and excludes it from Eq. (8). This is necessary for exact gain/loss null sectors, but a finite rapidity also obeys $a_j = \tanh(t\epsilon_j)$ and therefore approaches an endpoint exponentially. The fixed tolerance implies an approximate dynamic-range ceiling

$$\epsilon_{\text{max}}^{\text{resolvable}}(t) \simeq \frac{\text{artanh}(1 - 10^{-8})}{t} \simeq \frac{9.56}{t}. \quad (9)$$

Going to longer time can therefore make this estimator *less* informative in ordinary double precision.

The tangent and Choi calculations are different transfer objects; no exact identity between their finite-time spectra is assumed. Agreement is valuable cross-validation that the two constructions see a common slow mode. Disagreement can arise from finite time, numerical conditioning, or genuinely distinct sector content.

E. What “Fock rank” means in these files

For a free-fermion transfer problem, a many-body excitation can be an allowed sum of single-particle rapidities. The postprocessor defines a rank- R proxy by summing the R smallest positive magnitudes. Every completed production campaign was run with `fock_rank=1`; consequently the saved “Fock” gap is exactly the one-body proxy. The ranks $R = 2, \dots, 10$ were generated later from the same $T/L = 2$ spectra. No separate fixed-charge, parity, particle-hole, or operator-sector transfer calculation was performed.

III. WHAT WAS SIMULATED

A. Geometry and circuit protocol

All production data use $N_x = 20$ and periodic length $N_y = L$. The domain-wall slab contains the columns $x = 4, \dots, 16$, thirteen columns in total, so the reduced one-body dimension is $2 \times 13L = 26L$. The CPU class is configured with a domain wall, one shell, $\alpha_1 = 1$, $\alpha_2 = 30$, and domain-wall truncation. The initial state is a random pure half-filled Gaussian covariance; the exterior is projected once into an onsite product state.

Each cycle visits the active slab in `raster_y` order. At each site the four measurement channels A_+, A_-, B_+, B_- are sampled by their Born probabilities. The upper and lower targets are respectively empty and occupied. The campaigns have no forced postselection and use perfect correction, so the accumulated record weight is the product of measurement Born probabilities. Tangent, Choi, and trajectory-weight observers are attached to the same canonical call to `classAU1FGTN.run_markov_circuit`; parallelism is across independent one-sample workers, with one BLAS thread per worker.

The analysis sets $\alpha = 1$. This is an anisotropy convention in Eqs. (2)–(6), not the mass parameter α_1 . No independent velocity or aspect-ratio calibration was performed, so any quoted c_{eff} or x is conditional on this convention.

B. Saved estimators and uncertainty convention

For each size the production script saves the final cumulative mean of $Y/(\alpha Lt)$ and a 512-resample trajectory bootstrap standard error. It saves the median smallest finite tangent and Choi rate and a corresponding bootstrap standard error. The campaign fit is an unweighted two-parameter least-squares fit to the central values. Its reported slope error comes only from the residual scatter of two or three points and does not propagate the per-size bootstrap errors. Throughout this note those tiny three-point errors are described as regression diagnostics, not as full physical uncertainties. Weighted fits using the saved per-size errors and through-origin fits are quoted as sensitivity tests.

There is no burn-in: tangent rates and the cumulative free energy begin at the first cycle. The convergence diagnostic used below compares the mean value over the final quarter of the time series with the mean over the preceding quarter,

$$\delta_Q = 100 \left(\frac{\overline{O}_{[3T/4, T]}}{\overline{O}_{[T/2, 3T/4]}} - 1 \right) \%. \quad (10)$$

C. Campaign ledger

IV. NUMERICAL RESULTS

A. Free energy and effective central charge

The central result is instability, not a central charge. Table II lists the free-intercept fits used by the repository and weighted sensitivity fits. Increasing T/L changes both sign and magnitude of the result. The $T/L = 20$ sizes also span only $20 \leq L \leq 24$, making the intercept and L^{-2} slope particularly covariant.

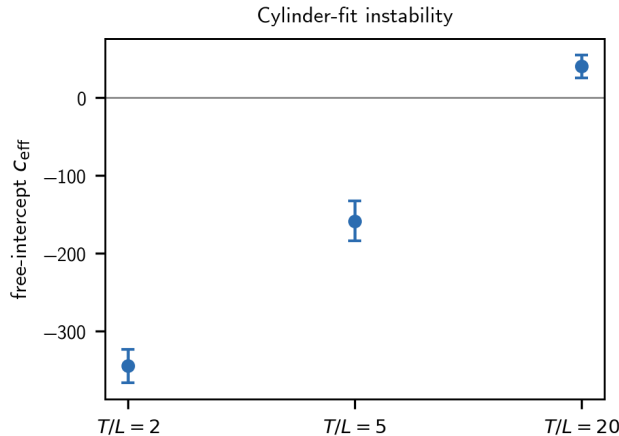


FIG. 1. The nominal cylinder-fit central charge is not stable under increasing T/L . Bars show only the unweighted three-point regression diagnostic.

The free-energy quarter drifts are $(-7.65, -5.71, -4.45)\%$ at $T/L = 2$, $(-3.61, -2.80, -2.18)\%$ at $T/L = 5$, and $(-0.80, -0.91, -1.01)\%$ at $T/L = 20$ in ascending L . The longer run is visibly better, but a one-percent temporal bias is still large compared with the desired cylinder correction. For orientation, a $c = 1$ signal changes f_0 by only about 4×10^{-4} between $L = 20$ and $L = 24$, whereas the standard error of their observed difference is about 1.4×10^{-2} . Ten trajectories are therefore underpowered for a unit-order central charge even before accounting for drift.

As an explicit estimator sensitivity check, fitting the slope of Y/L over the final half of each $T/L = 20$ trajectory gives stationary-rate estimates approximately 3.170 ± 0.019 , 3.171 ± 0.021 , and 3.175 ± 0.008 for $L = 20, 22, 24$. Their weighted cylinder fit yields $c_{\text{eff}} = 13 \pm 50$. This supports a bulk Born free-energy rate near 3.17 and, correctly, no useful central-charge determination.

B. Gap and scaling-dimension estimates

The direct size estimator

$$x_{\text{dir}}(L) = \frac{L\Delta(L)}{2\pi\alpha} \quad (11)$$

is the least extrapolation-dependent view of the gap data. Figure 2 shows that the $T/L = 5$ tangent and Choi points lie near one another and near 0.19, whereas the long tangent values fall across 0.281, 0.276, and 0.237. The latter trend is not a stable plateau.

The moderate-time result is the most internally coherent spectral observation. The Choi free-intercept is 2.60×10^{-5} , consistent with zero at the three-point regression level, and both through-origin routes yield $x \simeq 0.192$. However, the tangent intercept is 2.04×10^{-4} , the final-quarter drifts are still $(18.2, 15.8, -3.0)\%$ for tangent and $(31.4, 12.7, 6.3)\%$ for Choi, and the per-size uncertainties broaden the two values to 0.168 ± 0.032 and 0.186 ± 0.028 . The correct statement is therefore that $x \approx 0.19$ is a cross-validated finite-time candidate.

The long tangent free-intercept fit illustrates the danger of extrapolating three narrow sizes: it gives $x = 0.3785$ while its fitted intercept is $-1.47(66) \times 10^{-3}$. A through-origin fit gives 0.2696, close to the mean of the three direct estimates, 0.2646. This 40% shift is much larger than the displayed regression error. The long tangent quarter drifts remain 2.1–3.3%. It is premature to decide whether the asymptotic value is near 0.19, near 0.25, or elsewhere.

TABLE I. Completed production campaigns used in this note. Each size contains ten trajectories.

label	L	cycles T	T/L	role
short	20, 30, 40	40, 60, 80	2	pipeline and rank scan
moderate	20, 30, 40	100, 150, 200	5	best tangent–Choi overlap
long	20, 22, 24	400, 440, 480	20	temporal convergence stress test

TABLE II. Central-charge fits from completed campaigns. “OLS diagnostic” is the saved unweighted three-point regression error; “WLS sensitivity” propagates the saved per-size standard errors but still does not include temporal or window systematics.

T/L	sizes	c_{eff} (OLS diagnostic)	c_{eff} (WLS sensitivity)
2	20, 30, 40	-345.17 ± 21.48	-347.85 ± 35.16
5	20, 30, 40	-158.55 ± 25.38	-165.45 ± 29.83
20	20, 22, 24	$+40.16 \pm 15.02$	$+45.29 \pm 34.98$

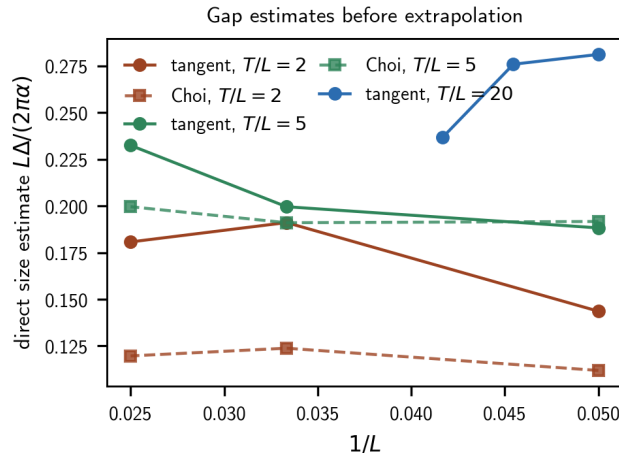


FIG. 2. Direct, per-size scaling estimates. These make visible the finite-time evolution that is hidden by a single extrapolated headline. No Choi point exists at the final time of the $T/L = 20$ run.

C. Why the long-time Choi gaps disappear

At $T/L = 20$ the final reduced Choi dimension is entirely classified as endpoint: 520, 572, and 624 modes for $L = 20, 22, 24$. Every one of the ten samples at every size has a nonfinite final Choi gap. The last cycles at which *any* sample retains a finite gap are 161, 191, and 259, corresponding to $t/L \simeq 8.05, 8.68$, and 10.79. Across the full histories only 1294/4000, 1511/4400, and 1846/4800 sample-cycle gaps are finite.

The saved endpoint flag is therefore a finite-precision statement at long time, not a count of exact null sectors. Moreover, `choi_censored_count=0` does not contradict the failure: the “censor” branch regularizes small denominators without deactivating the sample, the recorder leaves the active flag unchanged, and final aggregation does not use that flag to filter gaps. The final Choi eigenvalues lie within roughly 10^{-11} of the endpoints, far inside the 10^{-8} threshold. Shorter-time Choi gaps should consequently be treated as finite-time, finite-dynamic-range estimates, not a demonstrably converged limit.

D. Additive rank scan

The $T/L = 2$ postprocessing scan gives tangent/Choi pairs

$$(x_1, \dots, x_5)_{\text{tan}} = (0.126, 0.343, 0.684, 1.246, 1.935), \quad (12)$$

and

$$(x_1, \dots, x_5)_{\text{Choi}} = (0.108, 0.356, 0.713, 1.230, 1.937). \quad (13)$$

The agreement for ranks 2–5 is at the few-percent level. Choi support then collapses: rank five has only 7, 7, 8 valid samples across the sizes, rank six has 3, 0, 3, and higher ranks have none. The tangent construction continues through rank ten.

This is strong evidence that both Gaussian routes order the same low-lying one-body content before Choi saturation. It is not evidence for particular CFT primaries: additive order statistics automatically produce a smooth convex tower, and the short-time spectra are the least converged data in the campaign.

TABLE III. Scaling-dimension fit sensitivity. The free-intercept uncertainty is only the three-point regression diagnostic; the WLS uncertainty uses saved gap errors.

T/L	route	free-intercept x	WLS x	through-origin x
2	tangent	0.1256 ± 0.0203	0.1556 ± 0.0435	0.1529
2	Choi	0.10775 ± 0.00551	0.1130 ± 0.0388	0.1141
5	tangent	0.17475 ± 0.00450	0.1676 ± 0.0321	0.1922
5	Choi	0.18974 ± 0.00251	0.1863 ± 0.0281	0.1920
20	tangent	0.3785 ± 0.0496	0.3645 ± 0.0199	0.2696
20	Choi	no finite final-time rapidity		

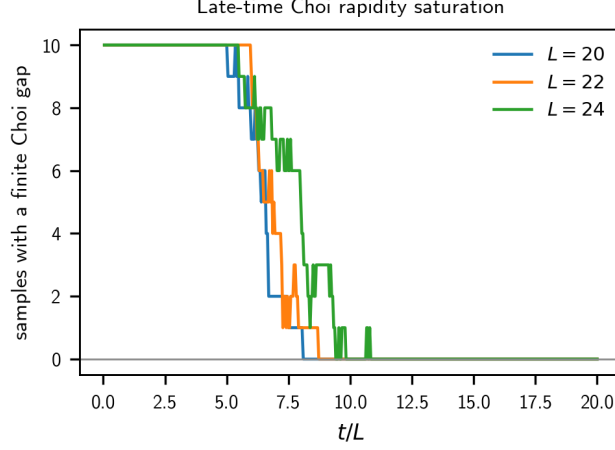


FIG. 3. Number of samples, out of ten, with a finite Choi gap during the completed $T/L = 20$ campaign. Finite rapidities are eventually mapped inside the fixed 10^{-8} endpoint tolerance and disappear.

V. IMPLEMENTATION AND PROVENANCE AUDIT

A. Reproducibility hole

The driver hashes the declared seed and sample index, calls `numpy.random.seed(seed)`, and records the rule in the manifest. It does not pass that seed as `random.seed` to `run_markov_circuit`. The default initial state inside the engine uses a new `numpy.random.default_rng()` seeded from entropy. Thus the measurement draws may follow the legacy global seed while the initial random pure covariance does not. Repeating the same declared sample does not reproduce the full trajectory, and nominally overlapping short and long runs are not exact prefixes.

This does not invalidate the saved samples as stochastic draws, but it prevents paired finite-time comparisons and makes the manifest’s seed rule incomplete. It also means that the separately completed $T/L = 5$ sizes can be combined only as independent samples under a common declared ensemble, not as a prefix-coupled dataset.

B. Manifest completeness and code state

The manifests record the repository commit and principal sweep flags but omit fixed settings needed to reconstruct the ensemble: domain-wall activation, shell count, mass parameters, truncation, initialization mode and filling, site sequence, active-slab restriction, and the physical-update convention. In addition, the extraction directory was untracked and the engine had local modifications, so the recorded commit alone does not pin the executed code. The long live process also loaded its Python source before a later source-file modification. Future manifests should record a source-tree diff or content hashes for the driver and engine, plus the complete resolved configuration.

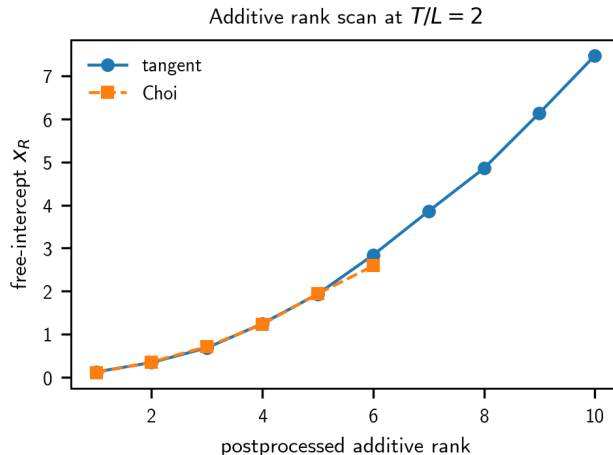


FIG. 4. Free-intercept fits to additive sums of the R smallest one-body rates at $T/L = 2$. The smooth, convex sequence is a real structural observation, but it is neither an independently simulated many-body tower nor sector resolved.

C. What the tests establish

The local suite contains eight passing checks covering synthetic fit algebra, endpoint classification, gap helpers, observer plumbing, refusal of postselection, parallel output shapes, and a small benchmark. The tangent primitive has additional cocycle tests elsewhere in the repository. These checks establish that the pipeline is wired together and that the saved quantities follow the intended elementary formulas.

They do not test finite-time convergence, full seed reproducibility, late Choi saturation, regularization bookkeeping, inactive-sample filtering, tangent null masking, a nontrivial Fock sector, or an exactly solvable tangent–Choi spectral comparison. Scientific convergence therefore cannot be inferred from unit-test success.

VI. INTERPRETATION

A. What is supported

The most defensible physical statement is that this interface circuit has a slow transfer channel whose finite-time rate scales roughly as $1/L$ over the simulated window. Two substantially different Gaussian diagnostics identify the same channel at moderate time, their trajectory-level values are strongly correlated, and their additive low-lying order statistics agree before the Choi representation runs out of precision. These facts make it unlikely that the entire signal is a plotting or indexing artifact.

A second supported statement is that the stationary bulk Born free-energy density is near 3.17 in the long runs. Its small residual size dependence is presently statistically consistent with many values of c_{eff} , including an ordinary order-one value.

B. What is not supported

The experiments do not support a numerical value or even a stable sign for c_{eff} . The large negative short-time fits should not be called exotic central charges; they follow the same direction as the explicit temporal drift and change radically with run length. Nor do the experiments select a final x^{typ} . The value near 0.19 is the best cross-route finite-time candidate, while the longer tangent data suggest a larger direct scale but fail the extrapolation-stability test. The Choi route cannot arbitrate because it has saturated.

The endpoint count is not a universal observable in its current form. It is dominated by the extensive reduced dimension and, at long time, by tolerance-dependent numerical saturation. The rank scan is not a conformal tower assignment because sector selection is absent. Finally, the present data cannot test whether $\alpha = 1$ is the isotropic CFT normalization.

C. Overall take

These experiments are a successful stress test of the proposed extraction strategy and a failed asymptotic measurement. That distinction is useful. The data identify precisely where the naive program breaks: cumulative free energies converge too slowly to resolve an L^{-2} Casimir term at ten samples; free-intercept gap fits are fragile over three sizes; and the direct Choi parametrization loses exponential information at the times needed for Lyapunov convergence. The correct next step is not to quote a compromise number from the existing fits, but to redesign the estimators around those failure modes.

VII. RECOMMENDED NEXT NUMERICAL PROGRAM

1. **Repair reproducibility first.** Pass the resolved sample seed directly to the canonical engine, record it, and test bitwise prefix equality across run lengths and worker counts. Save content hashes for all executed sources and the fully resolved model settings.
2. **Separate the cheap free-energy campaign.** Run a Born-weight-only ensemble with many more trajectories and more sizes. Estimate $f_0(L)$ from late-time slopes or blocked increments, scan the fit window, and use a nested trajectory bootstrap for the finite-size fit. Compare L^{-2} and $L^{-2} + L^{-4}$ models. Suppress a central-charge headline unless temporal drift is below the targeted Casimir signal.
3. **Use stationary tangent windows.** Introduce a burn-in, accumulate QR rates in late windows, and verify window stability. Report direct $L\Delta/(2\pi\alpha)$ points, free-intercept fits, and through-origin fits together. Increase the L range before interpreting a fitted intercept.
4. **Reformulate the Choi extraction.** Carry logarithmic singular values or a QR/SVD product representation rather than recovering rapidities from $a = \tanh(t\epsilon)$ near $|a| = 1$. At minimum save finite-mode counts, last finite cycle, denominator regularization records, and tolerance scans. A precision and endpoint-tolerance study is a required validation, not an optional cosmetic check.
5. **Define physical sectors.** Construct allowed pair or particle-hole excitations at fixed charge/parity and compare them with an independently measured correlator. Do not label sums of the R smallest magnitudes as CFT operators without this selection rule.
6. **Calibrate geometry.** Test N_x convergence or scale the transverse width with L , and determine the velocity/anisotropy factor α . Universal dimensions are otherwise quoted in an arbitrary temporal normalization.
7. **Build automatic publication gates.** Refuse a headline fit when fewer than three sizes survive, a convergence flag fails, the free intercept is significant, the Choi support count is inadequate, or a rank is not supported across every size. Store weighted, unweighted, through-origin, window, and bootstrap sensitivities in the output rather than only in a notebook.

Appendix A: Completed scalar values

The completed final scalar values are collected in Table IV; no row from the live extension is present.

TABLE IV. Final cumulative free-energy and rank-one gap values read from the completed scalar files. Parentheses show saved per-size bootstrap standard errors.

T/L	L	f_0	Δ_{tan}	$x_{\text{tan}}^{\text{dir}}$	Δ_{Choi}	$x_{\text{Choi}}^{\text{dir}}$
2	20	3.856(25)	0.045(12)	0.1436	0.035(9)	0.1119
2	30	3.620(35)	0.0400(65)	0.1911	0.0259(49)	0.1238
2	40	3.511(24)	0.0284(27)	0.1807	0.0188(60)	0.1197
5	20	3.438(25)	0.0591(90)	0.1882	0.0602(74)	0.1917
5	30	3.340(20)	0.0418(38)	0.1996	0.0400(32)	0.1910
5	40	3.275(15)	0.0365(34)	0.2324	0.0314(36)	0.1996
20	20	3.227(12)	0.0883(12)	0.2812	—	—
20	22	3.232(7)	0.0788(11)	0.2759	—	—
20	24	3.244(8)	0.0619(19)	0.2366	—	—

Appendix B: Artifact map and reproducibility of this audit

The raw roots are:

- `outputs/tmux_cft_20260806_000032/production/20260806_001959_88d7bf97e4;`
- `outputs/tmux_L20_T100_20260807_183409/production/20260807_183409_6ea93098a1;`
- `outputs/tmux_L30L40_T5L_20260808_133322/production/20260808_133322_0188a5388e;`
- `outputs/tmux_Nx20_L20_22_24_TNxNy_20260810_161330/production/20260810_161331_bcda51e8d2.`

All paths are relative to `cpu.cft.extraction`. The script `docs/build_experiment_note_assets.py` validates the completed time ratios, combines only compatible scalar rows, recomputes OLS/WLS/through-origin fits and quarter drifts, evaluates late-half free-energy slopes and Choi support histories, and writes `docs/generated/audit_summary.json`, `docs/generated/completed_campaign_rows.csv`, and the figures used here. The JSON also records that the live extension was excluded. Running the script never invokes the dynamics engine and never writes into a production directory.